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DEPARTMENT OF COMPUTER SCIENCE ENGINEERING

MAJOR PROJECT

ACADEMIC YEAR : 2026-2027

CLASS : IV Year CSE

Title of project: Hybrid Educational Chatbot integrating AIML 2.0 and Ontology based Reasoning for Intelligent Tutoring

ABSTRACT

Conversational learning platforms are becoming increasingly important in modern education, but traditional chatbots often struggle with rigid responses, limited context awareness, and a lack of subject-specific understanding. This project presents a Hybrid Educational Chatbot that combines the AIML-based CHARLIE system with the advancements found in AIML 2.0 and ChatScript to deliver more natural and context-aware conversations. By integrating ontology-based reasoning, the chatbot can understand relationships between concepts and provide accurate, subject-relevant responses instead of relying purely on pattern matching.

The system supports dynamic learning by adapting to student queries in real time, improves natural language interaction, and ensures mobile compatibility so learners can access tutoring support from any device. This makes the chatbot a flexible and intelligent assistant for learning platforms, capable of scaling across different subjects and educational levels. The proposed hybrid approach reduces the dependency on manual tutoring for repetitive queries, offers students instant and personalized assistance, and enhances the overall learning experience.

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